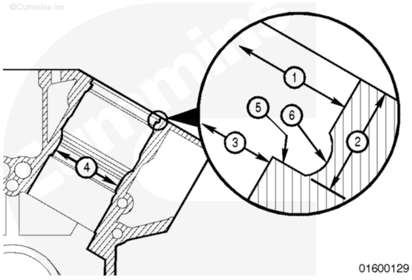
# Trainings

# General Information

Definition of numerical callouts:

1. Upper counterbore inside diameter
2. Counterbore depth
3. Lower counterbore inside diameter
4. Packing ring bore
5. Counterbore ledge
6. Counterbore radius.

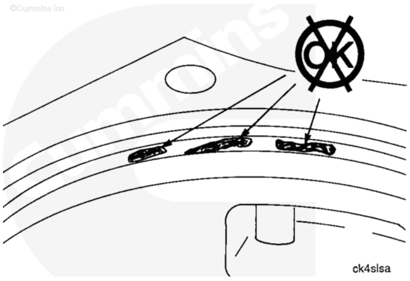



# Inspect for Reuse

Check the liner seat of the cylinder block for pitting.

Pitting on the liner seat is **not** acceptable.

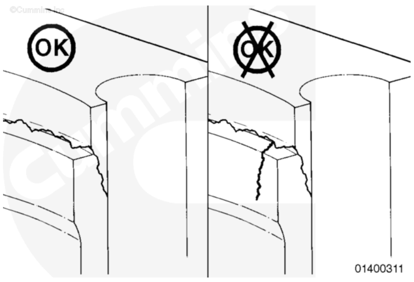
If the cylinder bore is pitted, the counterbore **must** be machined before the cylinder block can be used.




Use crack detection kit, Part Number 3375432 or equivalent, to check the counterbore ledge for cracks.

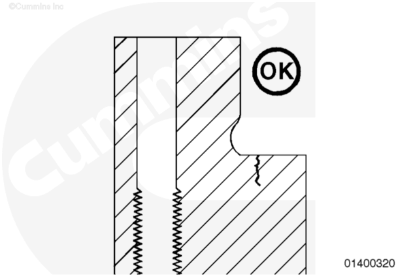
Circumferential cracks of the counterbore ledge are acceptable if the cracks do **not** extend to, or over, the edge of the ledge, as shown in the illustration.

Circumferential cracks in the radius are acceptable if they do **not** extend more than 90 degrees around the circumference of the counterbore radius.



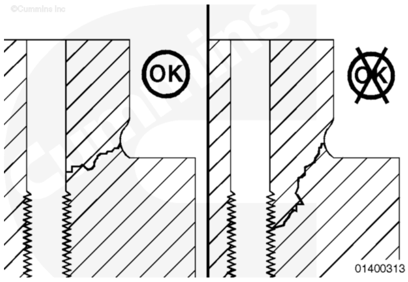

Metallurgical analysis of cross sections of counterbores having circumferential cracks has revealed that the cracks are surface initiated on the top of the counterbore ledge and normally do **not** propagate vertically through the counterbore ledge into the coolant passage around the cylinder liner.

Circumferential cracks in the radius are acceptable if they do **not** extend more than 90 degrees around the circumference of the counterbore radius.



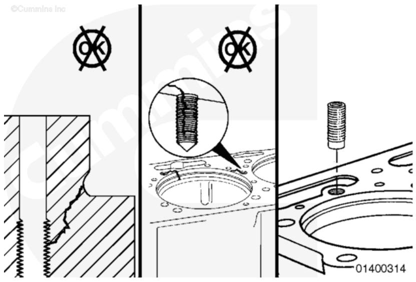

Check the capscrew holes for cracks.

Cracks that extend from the counterbore wall to the capscrew hole are acceptable for use **only** if they do **not** extend into the threaded portion of the hole.



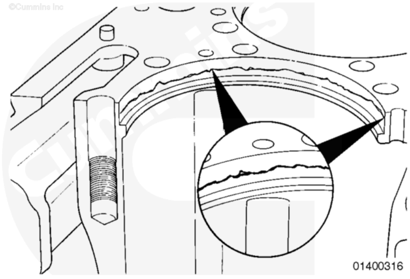

Cracks that extend into the threaded portion of the capscrew hole require repair with a blind-end thread insert.

Reference Section 11 (Alternative Repair Parts Group) of the Service Products Catalog, Bulletin 3377710, for a list of available thread inserts.



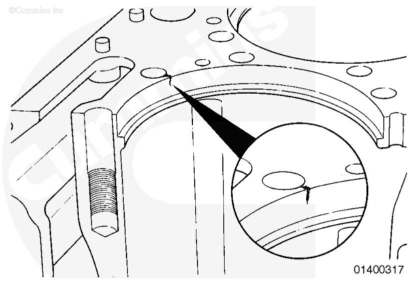

Check for cracks running horizontally around the vertical wall of the counterbore.

All cracked coolant passages closest to the bore **must** be repaired with coolant passage threaded inserts.

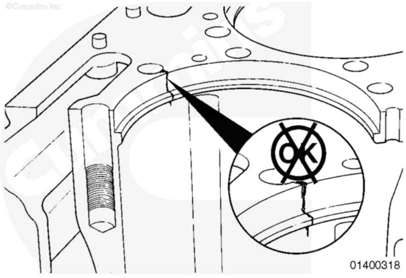



Check for cracks that run vertically to a coolant passage or a capscrew hole.

These passages **must** be repaired with coolant passage threaded inserts.




Cylinder blocks with vertical cracks that extend from a coolant passage down over the counterbore ledge can **not** be repaired.



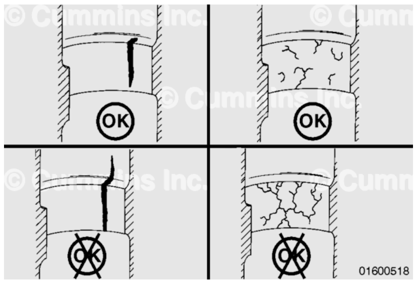

Packing ring bores that have up to six shallow fractures, which have random origins, **must** be repaired.

Fractures that extend across the bore and are less than 4.585 mm [ 0.181 in ] deep **must** be repaired.

**Note : This is the largest depth which can be repaired with the use of Cummins® repair sleeves.**

Reference the Machine section below.

**Note : Any fractures that extend into the coolant passage can not be repaired.**

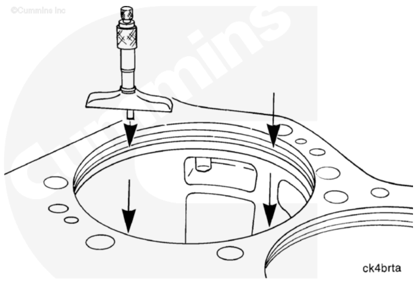



## Measure

Use a depth micrometer to measure the counterbore depth at four places 90 degrees apart, as illustrated.

The micrometer **must** contact the flat surface of the ledge and **must not** contact the radius.

If the counterbore depth does **not** meet specifications, a thicker or thinner seal ring was possibly used in a previous repair process. Measure and record the counterbore depth so that the liner protrusion can be determined.



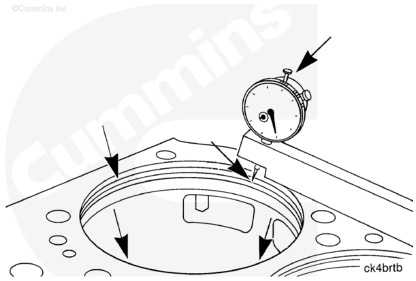

| Counterbore Depth | | |
|---|---|---|
| mm | | in |
| 13.684 | MIN | 0.539 |
| 13.734 | MAX | 0.541 |

The four measurements **must not** vary more than 0.055 mm [ 0.0021 in ]. If the measurements exceed the specifications, the counterbore ledge **must** be machined.

Use depth gauge assembly, Part Number 3823495, to measure the angle of the counterbore ledge at four places on the counterbore circumference.

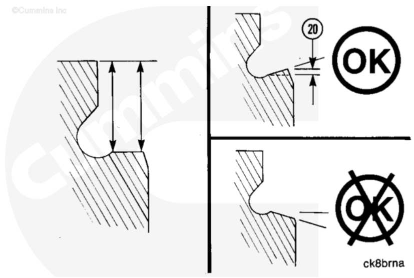
The indicator **must not** contact the counterbore radius on a block that does **not** have a double undercut.

The measurement of the ledge depth **must** be as near to the counterbore radius as possible, and as near to the counterbore edge as possible.




The angle (20) of the counterbore ledge is acceptable if the measurement that is near the counterbore edge is the same or no more than 0.036-mm [0.001-in] shorter than the measurement near the counterbore radius.

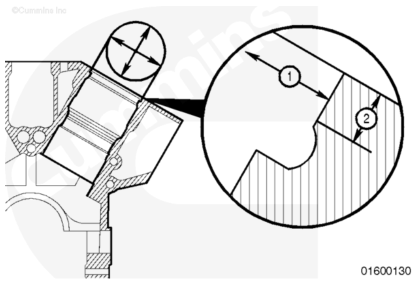
If the measurement near the counterbore ledge is longer than the measurement near the counterbore radius, the ledge **must** be machined. Reference the Machine section below.




Measure the inside diameter of the upper counterbore (1).

The point of measurement **must** be within 2.54 mm [ 0.100 in ] of the top of the block.

| Counterbore Diameter - Upper Press Fit Diameter | | |
|---|---|---|
| mm | | in |
| 190.28 | MIN | 7.491 |
| 190.34 | MAX | 7.494 |



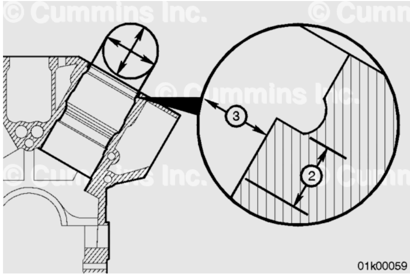

The inside diameter of the upper counterbore **must** be completely round within 0.050 mm [ 0.002 in ]. If the measurement is **not** within specification, check to see if the block can be machined to use oversize liners.

The upper counterbore **must** be no more than 0.076 mm [ 0.003 in ] larger than the cylinder liner flange.

If the counterbore does **not** meet specifications, an oversize liner was possibly used in a previous machining bore process. Measure and record the measurement so that the proper press fit liner can be determined.

Measure the inside diameter of the lower counterbore (3).

The point of measurement **must** be within 2.54 mm [ 0.100 in ] of the top of the counterbore ledge.



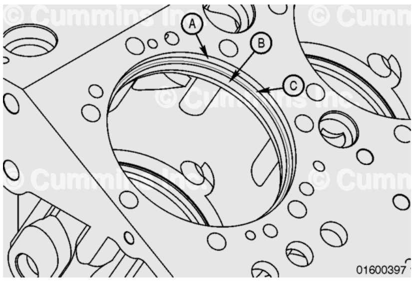

| Counterbore Diameter - Lower Press Fit Diameter | | |
|---|---|---|
| mm | | in |
| 181.74 | MIN | 7.155 |
| 181.80 | MAX | 7.157 |

The inside diameter of the lower counterbore **must** be completely round within 0.050 mm [0.002 in].

If the counterbore does **not** meet specifications, an oversize liner was possibly used in a previous machining bore process.

Measure and record the measurement so the proper press fit liner protrusion can be determined.

If the block is **not** within specifications, do **not** use the block.



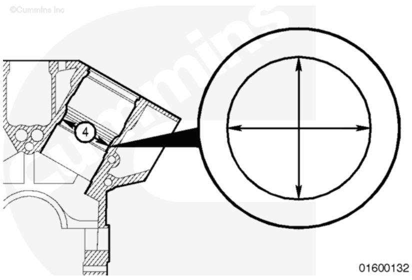

Two sizes of oversize liners have been released: 0.25 mm [ 0.0984 in ] and 0.50 mm [ 0.0197 in ] along with oversize liner shims for the respective oversize liners. The oversize dimensional features of the liner apply to the upper press fit diameter, lower press fit diameter **only**.

**Note : The 0.25 mm [ 0.0098 in ] oversize liner has the upper press fit dimension increased by 0.25 mm [ 0.0098 in ] from its nominal dimensional value.**

If an oversize liner is to be fitted, the engine block **must** be machined again for the respective oversize cylinder liner. The upper press fit (A) **must** be machined, or increased, to make sure of the correct fit of an oversize liner.

**Note : If a 0.25 mm [ 0.0098 in ] oversize liner is to be installed, the upper press fit dimension must be machined or increased by 0.25 mm [ 0.0098 in ] from its nominal dimensional value.**

Any number of cylinder bores can be machined for oversize liners.




Inspect the chamfer at the top of the packing ring bore.

Excessive pitting **must** be repaired. Reference the Machine section below.

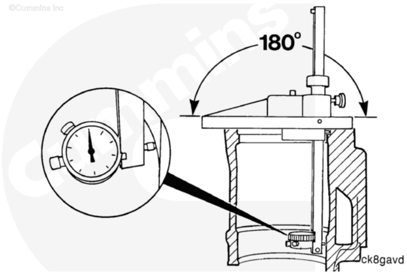
Measure the packing ring bore.

| Packing Ring Bore | | |
|---|---|---|
| mm | | in |
| 177.32 | MIN | 6.981 |

| Packing Ring Bore | | |
|---|---|---|
| mm | | in |
| 177.48 | MAX | 6.987 |

**⚠ CAUTION ⚠**

**Concentricity must be within specifications. Engine damage will result when the liner is not seated correctly in the bore.**

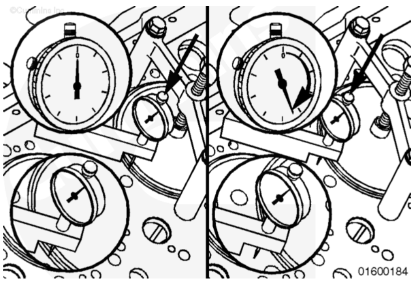



Use the concentricity gauge, Part Number ST-1252, to check the cylinder bore concentricity after the counterbore ledge has been repaired.

The bore runout is half of the indicator reading.

| Cylinder Block Counterbore Maximum Runout Value | | |
|---|---|---|
| mm | | in |
| 0.13 | MAX | 0.005 |

The bore **must** be repaired when consistent readings are **not** obtained. Reference the Machine section below.




Install a cylinder liner. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html)

Measure the cylinder liner protrusion. Refer to Procedure 001-064 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-064.html)

If the protrusion is **not** correct, remove the liner. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html)

The protrusion can be adjusted with seal rings or by machining the counterbore ledge. Reference the Machine section below.

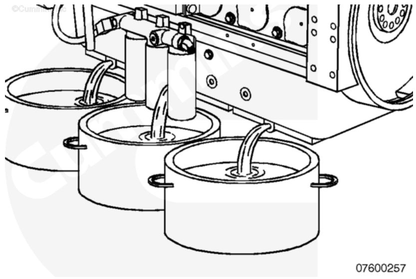
# Leak Test

**⚠ WARNING ⚠**

**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

**⚠ WARNING ⚠**

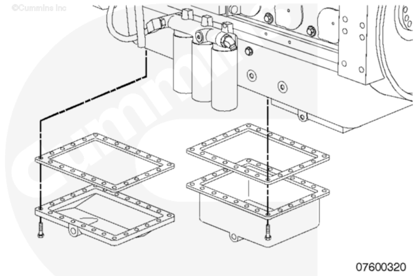
**Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**




Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)

**⚠ WARNING ⚠**

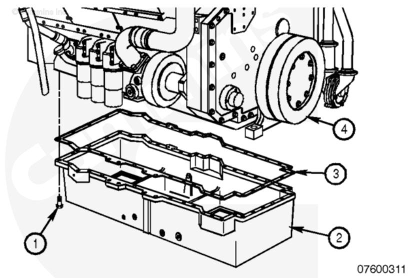
**This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**




Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)

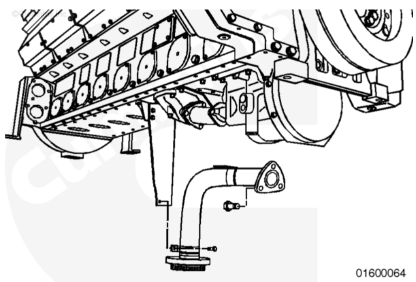
**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**



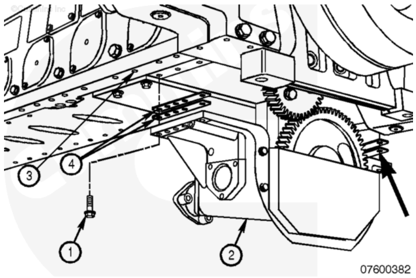

Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)

Remove the suction and transfer tubes. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)



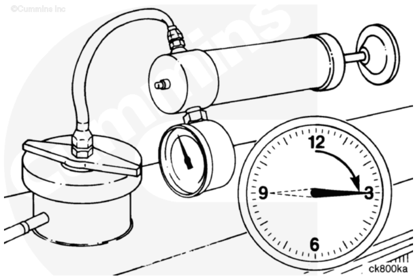

**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**




Remove the oil pump, if necessary. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)

Pressurize the engine cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)

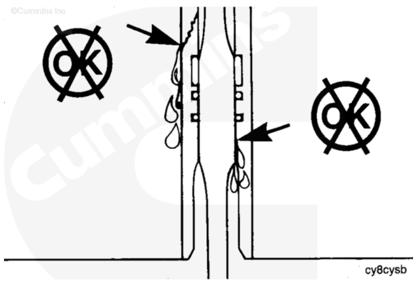



Apply the air pressure 15 minutes before inspecting the cylinder liner seats. Make sure the system is holding air pressure before beginning inspection.

**Measurements**

| | kpa | psi |
|---|---|---|
| Air Pressure: | 138 | 20 |

Inspect the outside diameter of the cylinder liners and the area below the cylinder liner seats in the cylinder block for coolant leaks.




If a leak is found, remove the cylinder liner(s).

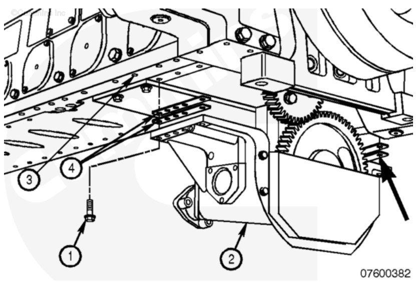
Inspect the sealing ring and cylinder liner. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html)

If a leak is found in the cylinder liner seat, reference the following procedure. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html)

Inspect the cylinder block liner counterbore area.

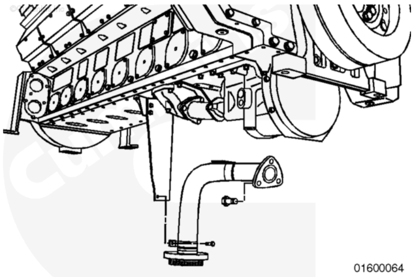
**⚠ WARNING ⚠**

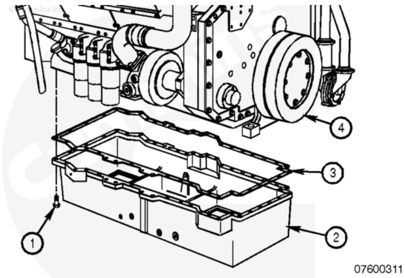
**This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**



Install the oil pump, if removed. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)

Install the oil suction and transfer tubes. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)

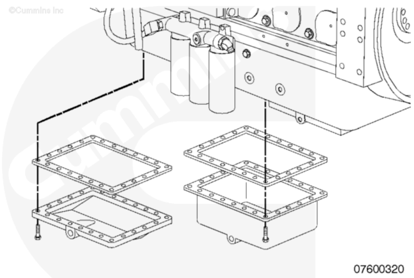



**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

Install the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)

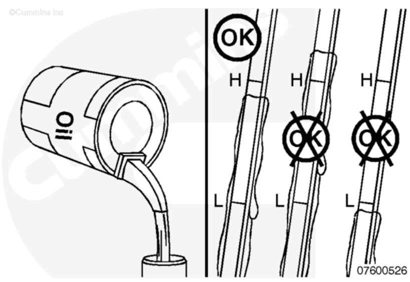



**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

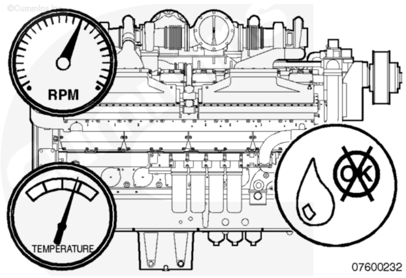
Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)

Fill the engine with clean engine oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)



Operate the engine until it reaches a temperature of 80°C [ 176°F ].

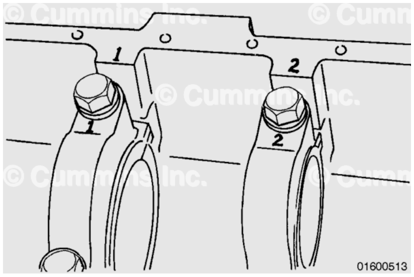
Check for coolant and lubricating oil leaks.




## Machine

**Note : If these repairs are done with the main bearing caps NOT installed correctly, the effectiveness of these repairs decreases.**

To machine the cylinder bores, first attach the main bearing caps. Refer to Procedure 001-016 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-016-tr.html)




The cylinder block head deck **must** be machined before the cylinder bores if the head deck requires repair.

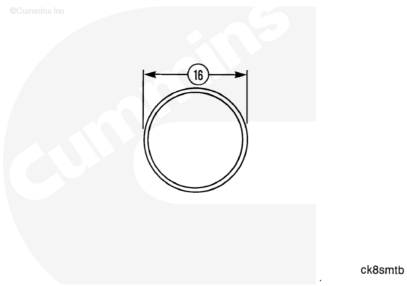
Use a capable machine shop to machine the cylinder bores.

### Counterbore

**Note : Below is the procedure for machining the upper counterbore.**

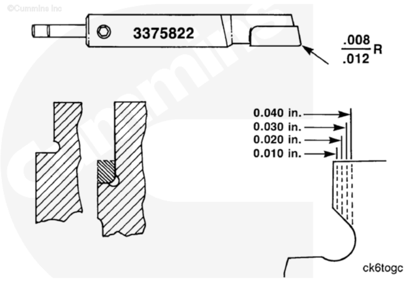
Select the appropriate repair sleeve from the upper deck repair bushings and the available service tooling, if required. Reference Section 11 (Alternative Repair Parts Group) of the Service Products Catalog, Bulletin 3377710.


Measure and record the outer diameter or the repair sleeve.



Machine the top liner bore to give interference and depth according to the table below.

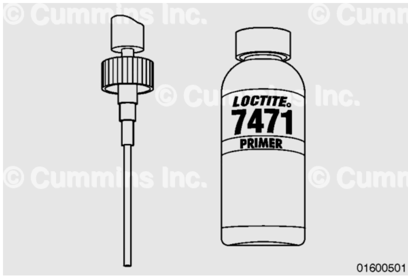
| "L" Shaped Sleeve Specifications (mm [ in ]) | |
|---|---|
| Interference Fit on Diameter | 0.064 to 0.076 [ 0.0025 to 0.0030 ] |
| Depth | 17.00 to 17.30 [ 0.669 to 0.681 ] |
| Corner Radius | 1.87 to 2.13 [ 0.074 to 0.085 ] |



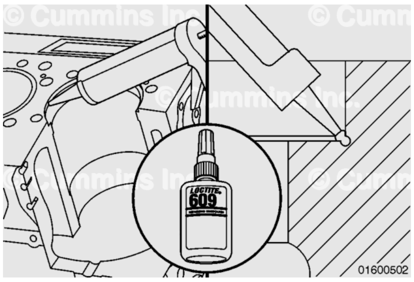

**⚠ WARNING ⚠**

**Use skin and eye protection when handling caustic solutions to reduce the possibility of personal injury.**

Thoroughly clean the outer diameter of the repair sleeve and cylinder block bore. Use cleaner/primer, Part Number 3824715.



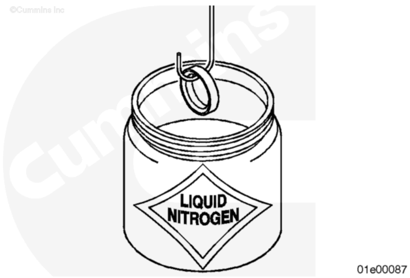

Apply retaining compound, Part Number 3823718 or equivalent, to the corner of the cylinder block counterbore.




Use liquid nitrogen to shrink the repair sleeve.

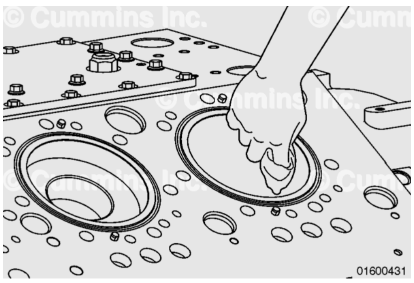
Install the sleeve into the cylinder block. Make sure the repair sleeve bottoms in the counterbore.

Hold in position until the repair sleeve expands enough to remain in the correct position in the cylinder bore.



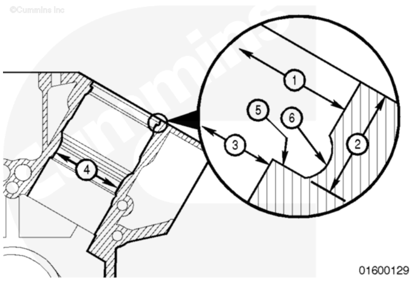

Remove all traces of surplus retaining compound from the cylinder block.

Allow 3 to 4 hours for the retaining compound to cure.



Machine the sleeve to the correct specifications as indicated in the measure section above, replacing the undercut (6) with a 1.00 - 0.80 mm [.040 - .030 in] corner radius.

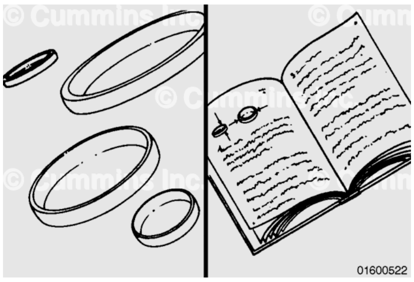
**Note : The sleeve will protrude above the head deck once installed and must be machined level with the deck head prior to machining the counterbore.**




**Note : Below is the procedure for machining the packing ring bore.**

Select a repair sleeve. Reference Lower Bore Repair Bushing/Tooling in Section 11 (Alternative Repair Parts Group) of the Service Products Catalog, Bulletin 3377710.

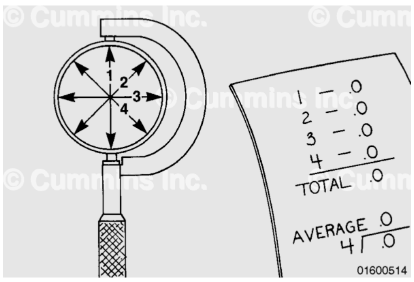
Cummins Inc. installation tooling is **not** available for this repair sleeve.




Measure the outside diameter of the repair sleeve in four equally spaced places.

Record these measurements.

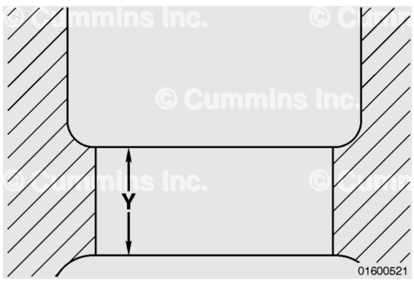
This value will be used to determine the correct size to cut the cylinder block.



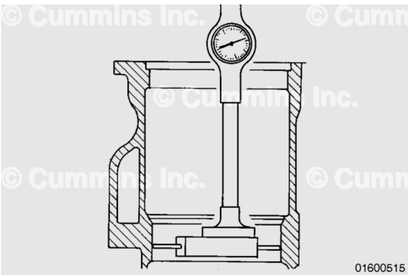

Machine the entire length of the lower packing bore area (Y) to provide the specified interference fit between the outer diameter of the repair sleeve and the cylinder block.

| Interference Fit on Diameter | | |
|---|---|---|
| mm | | in |
| 0.084 | MIN | 0.0033 |
| 0.159 | MAX | 0.0063 |

**Note : Removal of the liner catcher is required, as the sleeve will be installed from the bottom of the block.**

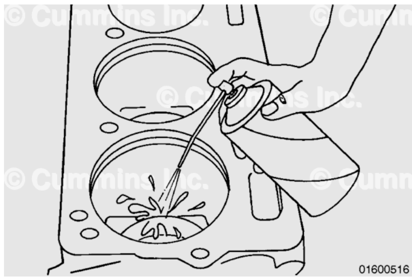



Measure the finished bore size to verify the inside diameter and distance from the cylinder block head deck surface are sized correctly.




# ⚠ WARNING ⚠

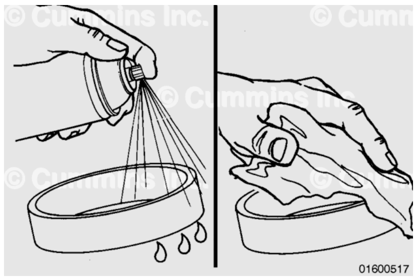
**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**




Thoroughly clean the bore surface area with spray cleaner, Part Number 3375433 or equivalent.

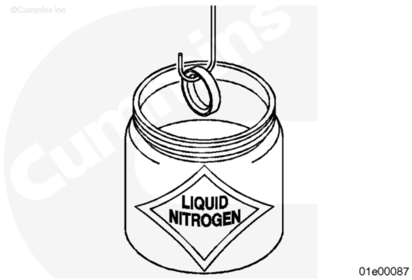
Dry with compressed air and wipe clean.

Clean the outside diameter of the repair sleeve with spray cleaner, Part Number 3375433 or equivalent.

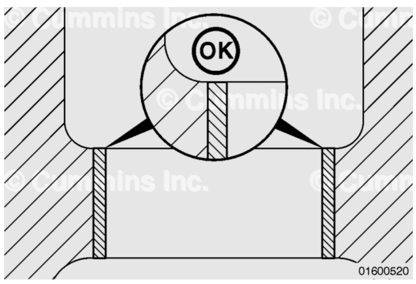



Use liquid nitrogen to shrink the repair sleeve.

Apply a 0.8 mm [ 0.03 in ] wide band of retaining compound Loctite™ 675, or equivalent, to the machined bore.

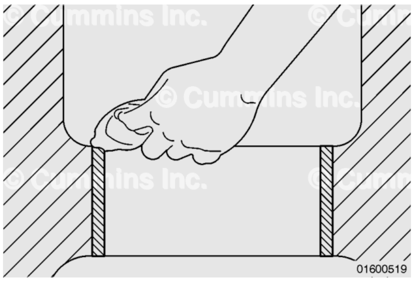



From the bottom of the engine, install the repair sleeve until the top edge is flush with the upper edge of the lower packing bore, as illustrated.



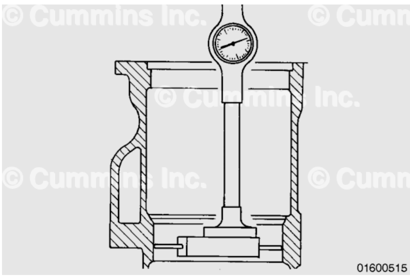
Wipe the excess Loctite™ compound from the cylinder block and upper portion of the cylinder sleeve.

Allow 3 to 4 hours for the retaining compound to cure.




Inspect all dimensions specified in the Measure section above.

Bore or hone the repair sleeve to the correct size, as required.




The cylinder block **must** be thoroughly cleaned prior to engine assembly.

For detailed cleaning instructions, reference the following procedure. Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-026.html)